BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Pg 21 / 22

CPU Performance

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. What is stored by the Program Counter (PC), and what is held in the Memory Data Register (MDR)?

2. Identify one job of the ALU and one job of the Control Unit.

3. Convert 4 KB to bytes using the decimal convention 1 KB = 1,000 bytes.

4. A 120 x 80 pixel image uses 8-bit colour depth. Calculate the uncompressed pixel-data size in bytes. Why should a source-code file use lossless rather than lossy compression?

Today's objectives

Identify clock speed, cache size and number of cores as the three CPU performance factors. Explain how changing each factor can affect CPU performance.

Key vocabulary - Read, cover, recall

Clock speed
The number of clock cycles the CPU carries out per second, measured in hertz.

Hertz (Hz)
Unit of clock speed: one clock cycle per second.

Cache
Small, fast memory that stores frequently used data and instructions for the CPU.

CPU core
Part of a CPU that processes instructions.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P1 / P2

Learn and practise

New learning - Four key ideas

CLOCK SPEED	Clock cycles each second, measured in Hz.
CACHE	Frequently used data/instructions near the CPU.
CORES	More cores can handle multiple tasks at once.
COMPARE	Use all three features; real speed depends on the task.

COMPARE THE SPECS, THEN EXPLAIN THE WORKLOAD

CPU C

3.6 GHz | 2 cores

4 MB cache

CPU D

3.0 GHz | 6 cores

12 MB cache

CPU D may suit several concurrent programs; no blanket winner.

I Do - Worked example

Using the CPU A and CPU B specifications shown, explain TWO reasons CPU B may be better for running several programs at the same time. [4 marks]

1. CPU A: higher clock speed. CPU B: more cores and cache.
2. More cores can run several programs concurrently.
3. More cache can reduce slower RAM accesses.
4. Explain each feature's benefit: two developed points.
5. Neither CPU is guaranteed fastest for all workloads.

We Do - Guided practice

Compare CPU C (3.6 GHz, 2 cores, 4 MB) and CPU D (3.0 GHz, 6 cores, 12 MB). Why may D help multitasking?

BLACK PEN - complete your explanation:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P1 / P2

Think and write

Code First - Predict > Write > Test > Improve

Read the OCR Exam Reference Language code before testing it.

```
clockJ = 2.6
clockK = 3.4
if clockJ > clockK then
print("CPU J has a higher clock speed")
else
print("CPU K has an equal or higher clock speed")
endif
```

1. Predict the exact clock-speed message.

2. Set clockJ = 3.8; predict the changed message.

3. Write an IF comparing coresJ=6 with coresK=4.

You Do - Independent exam questions

Q1. State the three characteristics of a CPU that affect performance. [3 marks]

Q2. A college chooses a CPU for computers running several applications.
CPU G: 3.9 GHz, 2 cores, 6 MB cache.
CPU H: 3.1 GHz, 8 cores, 16 MB cache.
Explain TWO reasons why CPU H could be better for this use. [4 marks]

Q3. Explain how clock speed, cache size and the number of cores each affect CPU performance. [6 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. A processor with more cores can improve performance when several programs are running. ☐ T ☐ F

2. Cache can reduce time accessing RAM. ☐ T ☐ F

3. More cores always make every single task run faster. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P1 / P2

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.

Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 PC: address of the next instruction to be fetched. MDR: the data or instruction currently being transferred to/from memory. Both are registers. ☐

Q2 ALU: arithmetic and logical operations. Control Unit: decodes instructions and coordinates/controls CPU activity. ☐

Q3 4 x 1,000 = 4,000 bytes. Include the unit. ☐

Q4 120 x 80 = 9,600 pixels. x 8 = 76,800 bits. / 8 = 9,600 bytes. Lossless is required to restore the original source code exactly. ☐

Code First - Answers

1. Initial: "CPU K has an equal or higher clock speed". ☐

2. After clockJ=3.8: "CPU J has a higher clock speed". ☐

3. With coresJ=6, coresK=4: "CPU J has more cores". ☐

We Do - Check the model

D has 6 vs 2 cores [2]: more tasks concurrently. 12 vs 4 MB cache [2]: more frequently used data nearby. C has the higher clock speed.

Exit Ticket - Check

1. TRUE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q1 Clock speed [1]; cache size [1]; number of cores [1]. ☐

Q2 H: 8 vs 2 cores -> better parallel tasks [2]; 16 vs 6 MB cache -> fewer slow RAM accesses [2]. ☐

Q3 Clock: cycles per second + faster processing [2]. Cache: fast frequent data + less RAM [2]. Cores: concurrent tasks + multitasking [2]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: